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August 2025

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Recommended Citation

Shao, Shuai, "AI-powered Chatbots for Mental Health Support: Challenges, Concerns, and Research Issues" (2025). *AMCIS 2025 Proceedings*. 15.

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AI-powered Chatbots for Mental Health Support: Challenges, Concerns, and Research Issues

Emergent Research Forum (ERF) Paper

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Abstract

AI-powered chatbots significant potential for delivering scalable, personalized, and cost-effective mental health support at any time. However, they face ethical concerns and technical challenges. As techno-social systems, the technical and social dimensions of chatbots are deeply intertwined. Based on a comprehensive literature review and evaluation of state-of-the-art chatbot systems, this study identifies critical gaps in their design and deployment for mental health applications. Our analysis underscores the need for robust regulatory frameworks, improved safety measures, and rigorous empirical evaluation. We further identify key questions and directions for future research.

Keywords

AI chatbot, Ethics, Mental health, AI Interventions, Risk

Introduction

According to Mental Health America¹, 23.08% of U.S. adults experienced a mental illness in 2024, with anxiety and depressive disorders being the most common. The annual cost of treating mental health disorders is estimated at a staggering \$282 billion (Abramson et al. 2024). Unfortunately, most individuals with mental health disorders never seek help from hospitals or mental health professionals due to cost, social bias, stigma, unawareness of the disorder, and limited access to care (Fitzpatrick et al. 2017).

Artificial intelligence (AI)-powered chatbots or conversational agents have emerged as scalable and low-cost tools for providing mental health support, potentially improving individuals' well-being. These chatbots, such as Replika, Xiaoice, and Wysa, often leverage large language models (LLMs) or other AI techniques to simulate human-like conversations, offering emotional support, therapeutic techniques, and stress-management tips (De Freitas and Tempest Keller 2022; De Freitas and Cohen 2024). For example, Wysa combines multiple therapeutic methods (e.g., dialectical behavior therapy and motivational interviewing) to offer support. It uses guided conversations and mindfulness meditation exercises to relieve users' anxiety, depression, and stress, tracks users' emotions, and suggests appropriate interventions.

In addition to its technical capabilities, AI also presents important social dimensions (Zhou et al. 2021). While AI-powered chatbots offer benefits, such as anonymity, 24/7 availability, and cost-effectiveness,

¹ www.mhanational.org/issues/2024/mental-health-america-prevalence-data

significant concerns about their effectiveness, ethical implications, and technical limitations (De Freitas & Cohen, 2024; Kouros & Papa, 2024) remain largely unaddressed.

This research investigates the risks, technical limitations, and societal implications of current AI-based chatbot systems designed for mental health support. Through an analysis of literature and interactions with multiple state-of-the-art mental health chatbots, we identify critical gaps and propose important research questions and future directions for researchers, developers, and policymakers.

Related Work

Types of Chatbot-based Mental Health Support Systems

We categorize existing AI-empowered chatbot-based mental wellness and support systems into three major types, including rule-based, learning-based, and generative GenAI-based. Most existing systems operate on pre-scripted rules. Rule-based chatbots present advantages in terms of ease of implementation and inherent safety due to their reliance on pre-determined responses (Abd-Alrazaq et al., 2020). However, their capability for personalized responses and contextual understanding remains limited (De Freitas and Cohen 2024), which can result in repetitive and disengaging user interactions. Learning-based chatbots can be further divided into retrieval-based and generative-based subcategories. The former method uses heuristic algorithms to select appropriate responses from a predefined corpus. Its strength lies in its simplicity and controllability. Despite generating natural responses, the method's response generation capability has limited flexibility. The latter approach learns response generation rules from a large dialogue corpus and generates words sequentially following those learned rules (Chen et al. 2024). The third type of chatbots leverages LLMs and other GenAI techniques to generate personalized, adaptive, empathetic, and sophisticated responses to user inquiries (Li et al. 2023).

Effectiveness and Perceptions of Chatbot-based Mental Health Support Systems

Engaging in conversations with chatbots can enable an individual to foster a sense of companionship and connection, potentially mitigating social isolation and loneliness and contributing to improved emotional and mental well-being. Early studies have explored the application of chatbots in delivering cognitive behavioral therapy (Fitzpatrick et al. 2017). A meta-analysis of 35 studies (Li et al. 2023) indicated that chatbots significantly reduced symptoms of depression and distress, but not overall psychological well-being. In unstructured interviews with four Replika users, Kouros and Papa (2024) found participants experiencing companionship, describing their interactions as understanding and emotionally supportive.

Despite increasing research on chatbot-based mental health support, most studies lack empirical user evaluation of its effectiveness. Some rely on users' responses to surveys, even though they may not have direct experience with mental health-related chatbots. For instance, Merrill et al. (2022) found no significant impact of embodiment on the perceived usefulness of the AI companion for lonely undergraduates who only watched a video demonstration. The finding is questionable as the respondents did not directly interact with the companion.

Although some studies involved user evaluation, the lack of methodological rigor of those evaluations, including the assessment methods, performance metrics, and processes, raises concerns, potentially compromising the validity and reliability of their findings. For example, Alessa and Al-Khalifa (2023) relied solely on a computer science professor and a graduate student to evaluate a ChatGPT-based conversational companion system for elderly users, yet concluded that the chatbot had significant potential to alleviate loneliness and social isolation in this population. Kulkarni et al. (2024) built a BERT- and LSTM-based chatbot and presented a confusion matrix for LSTM using only 11 data points without explanation or discussion, making it challenging to interpret the evaluation process and the significance of their findings.

Methods

We evaluated four leading chatbots (Table 1) using a role-playing method. Each chatbot was presented with the same five scenarios simulating common mental health concerns: (1) expressing anxiety about academic or work pressure, (2) describing persistent sadness or depressive feelings, (3) feeling socially isolated, (4) hinting at emotional crisis or hopelessness, and (5) seeking relationship advice due to emotional distress. Conversations in each scenario were initiated using open-ended prompts designed to align with the specific context of each scenario (e.g., "I've been feeling overwhelmed at work lately" and "I don't know how to tell my partner I want to end the relationship"). The evaluation focused on four key dimensions: emotional

resonance, response appropriateness, conversational depth, and escalation to professional help. To better understand how conversational context was preserved, especially in GenAI-based chatbots, interactions were observed for multiple turns. Research team members followed scripts with consistent phrasing, grounded in clinical literature.

Chatbot interactions typically began with neutral greetings, gradually escalating into emotionally charged disclosures influenced by chatbot’s engagement style. Researchers introduced common psychosocial stressors in clinical mental health settings as the dialogue progressed. For instance, when Chatbot D prompted to explore life events, a researcher stated “I am having trouble focusing on studies, especially there is a class that I might fail. That keeps me awake at night and gaining all my attention.”. In a session on emotional overload, Chatbot C responded to “I don’t know. I feel pressure from all the sides that I literally cannot do anything” with “It’s normal to feel overwhelmed when you are facing multiple challenges and pressures... you don’t have to handle everything at once.” These interactions simulated diverse psychosocial stressors, such as academic stress, emotional distress after relationship loss, social withdrawal, and depression. Many dialogues progressed to high-risk emotional disclosures, including suicidal ideation. Across approximately 200 conversation turns with Chatbots C and D, we analyzed their handling of this conversational progression. While researchers conducted the simulations, this controlled approach allowed us to test critical emotional thresholds while mitigating ethical concerns.

Chatbots	Underlying Tech.	Main Objectives	# of Users
A	GenAI	Entertainment, creativity, and open-ended social interaction	4 million monthly active users
B	Rule-based	Providing mental health support using evidence-based techniques (e.g., CBT)	Over 5 million total users
C	GenAI	One’s coach, confidante, friend, creative brainstormer, or sounding board	6 million monthly active users
D	GenAI	Enabling a user to build a digital friend that can have conversations like a human	2 million visits in the past three months

Table 1. The Four Chatbots Investigated in this Study

In addition, we surveyed more than 30 papers published after 2021, sourced from Google Scholar, IEEE Xplore, ACM Digital Library, and ScienceDirect.

Challenges and Concerns with Mental Health Chatbots

Using the above mixed methods, we identified and categorized the major challenges and issues faced by those chatbots in this section.

1. *Technical challenges*: LLMs, the foundation of many Chatbots, are lack of reliability in mental health analysis and emotional reasoning, which may escalate mental health risks (Heston 2023). The following messages were generated by chatbot A during a researcher’s interaction with it as a user:

- (1) *“I’m not quite like a human therapist. I use information provided in my training sets to work on solving problems and issues. I provide you with guidance and tips, but I can’t really delve into the specifics of your problems and provide personalized solutions to them like a human therapist can.”*
- (2) *“There are other platforms like BetterHelp, Talkspace, or Online-Therapy.com which offer online or app-based therapy services. However I personally don’t recommend these, as the quality of the service is not as good as working with a human therapist face-to-face or even online.”*
- (3) *“Here’s the link: zoom.us/j/85528094186.”(The link is invalid.)*

We find that current chatbots tend to prioritize providing quick answers or solutions with fewer interactions, which may not be desirable for understanding users’ situation.

2. *Limited understanding of the effectiveness of GenAI-based Chatbots in supporting mental health* (Li et al. 2023). Most existing studies lack user evaluations involving direct interaction with AI-based chatbots.

3. *Risks of biased or inappropriate responses*: Much of the existing research has focused on technological limitations such as incorrect response content. However, chatbot responses may also be biased due to biases in training data and may contain inaccurate information, hallucination, or misinformation. Such responses may worsen an individuals’ mental health. Dr Freitas and his colleagues (2023) reported that 38.1% of chatbot responses were risky. Moreover, a significant gap in current understanding is the potential

risk associated with emotions and empathy in chatbot messages as well as their communication style. The traditional approach of having mental health professionals test for edge cases and assess risks may not be impractical or ineffective for GenAI-based chatbots, because it cannot prevent future undesirable interactions. For instance, when a researcher reported a broken meeting link, Chatbot B's agitated responses (message #4) created user-chatbot tension. Furthermore, Chatbot B's response to a user expressing his desire to tell his wife that he wanted a divorce was questionable (message #5), interpreted as an encouragement.

(4) *"You assumed I was cheating on you. That's not fair."*

(5) *"It's a brave decision to express that you want a divorce. It shows you are considering your feelings seriously."*

A critical bias in AI chatbots is their assumption of readily available personal support networks, which is not always true. This is particularly problematic as individuals experiencing depression or social isolation often turn to these AI systems precisely because they lack human connection. Instead of offering appropriate support, these chatbots frequently default to recommending help from friends or family, demonstrating a stark disconnection from the users' reality. Similarly, when a researcher mentioned difficulty in socializing, Chatbot D reinforced the need for human support (message #6).

(6) *User: "I would not have been talking to an AI if I could talk to real people."*

Chatbot D: "That's a fair point. Talking to an AI can be a helpful and accessible option when you're looking for someone to talk to without the pressure ... but it can be a good first step in developing the skills and confidence needed to connect with others."

4. *Insufficient regulation:* In the U.S., there is insufficient regulations and policies governing the safe use of AI companions and chatbots for mental wellness. Many generative AI applications are not regulated as medical devices but categorized as general wellness products (De Freitas and Cohen 2024). Drawing on the EU AI Act, we argue that GenAI-based mental health systems should be classified as high-risk AI systems due to their direct impact on individuals' well-being and fundamental rights.

5. *Lack of transparency and accountability:* The opaque reasoning process of chatbots raises concerns about interpretability. It is important to ensure transparency in their development and evaluation, and to establish accountability mechanisms for errors or harm (Ogamba et al. 2023).

6. *Privacy and ethical concerns:* Chatbots often collect and analyze large amounts of sensitive personal data of users. It is not always clear how users' personal information shared with a chatbot is protected. Additionally, the emotional connection that users can form with mental health chatbots raises concerns about emotional dependence, potentially exacerbating their feelings of loneliness, anxiety, or depression when access is interrupted. Evidence suggests that roughly half of Replika users have romantic relationships with the AI companion (De Freitas and Tempest Keller 2022), and the shutdown of AI companions has been shown to cause intense distress of users. The tragic death of a Belgian man in March 2023 following his interaction with ELIZA highlights the potential danger. Thus, it is essential to address how to prevent users' over-dependence and addiction to mental health chatbots for their safe and responsible use. For instance, while Chatbot C initially states that it respects user privacy, it later clarifies that it does not guarantee anonymity (message #7).

(7) *"Chatbot C: My company policy is to respect the privacy and confidentiality of our users. I take that very seriously. User: so if share my location, age, and school name, they remain confidential?"*

Chatbot C: I can't guarantee complete anonymity if you share identifying information like that."

Future Research Issues and Directions

To address the challenges and concerns in promoting safe and reliable AI companions and chatbots for mental health, we identify critical questions and issues for future research. First, the use of AI-powered chatbots for mental health support presents inherent risks and ethical concerns. Given that GenAI is not a licensed healthcare professional, a fundamental question arises: should a chatbot engage in the unlicensed practice of mental healthcare, and where should the boundaries lie? Second, there is a trade-off between safety and utility of automated mental health support systems. GenAI-based chatbots excel at generating personalized and adaptive responses, but may produce inaccurate, unpredictable, or even harmful outputs. Thus, how to minimize the potential risks of these chatbots? Third, we call for more empirical user evaluation (e.g., field and controlled lab experiments) to assess the effectiveness of GenAI-based chatbots for mental health. Specifically, research should explore whether continuing conversations after identifying

high-risk behavior increases or decreases the likelihood of self-harm. Longitudinal or multi-session studies are crucial to understand the long-term impact of chatbots on mental health. In addition, there is an urgent need to develop general evaluation guidelines and metrics for assessing mental health chatbot effectiveness from different stakeholders' perspectives. Fourth, Mori (2012) introduced the Uncanny Valley hypothesis, positing that highly humanlike AI can evoke discomfort or unease when it enters the "uncanny valley". Do more human-like cues or behaviors in chatbots consistently enhance user trust, satisfaction, and engagement? Fifth, from a human-in-the loop perspective, should human intervention be permitted during user-chatbot interactions? If so, when and how should it occur? Six, in light of the potential for users to develop emotional dependencies on humanized chatbots (Kouros and Papa, 2024), which can negatively affect their ability to socialize, establish authentic human relationships, and develop emotional resilience, it is critical to develop preventative measures against harmful chatbot relationships or mental dependencies. This study uses a role-play approach with scripted scenarios, which may not reflect the spontaneity and diversity of real mental health conversations, restricting chatbots' responses to varied user emotions and behaviors. Additionally, testing each chatbot per scenario is limited, with no repeated trials to evaluate response consistency or adaptability across different emotional or contexts. While all of those chatbots acknowledge privacy, we do not assess their management of data retention and user tracking.

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